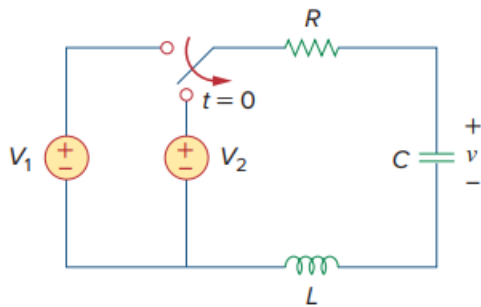


Problem

Using Fig. 8.83, design a problem to help other students better understand the step response of series RLC circuits.

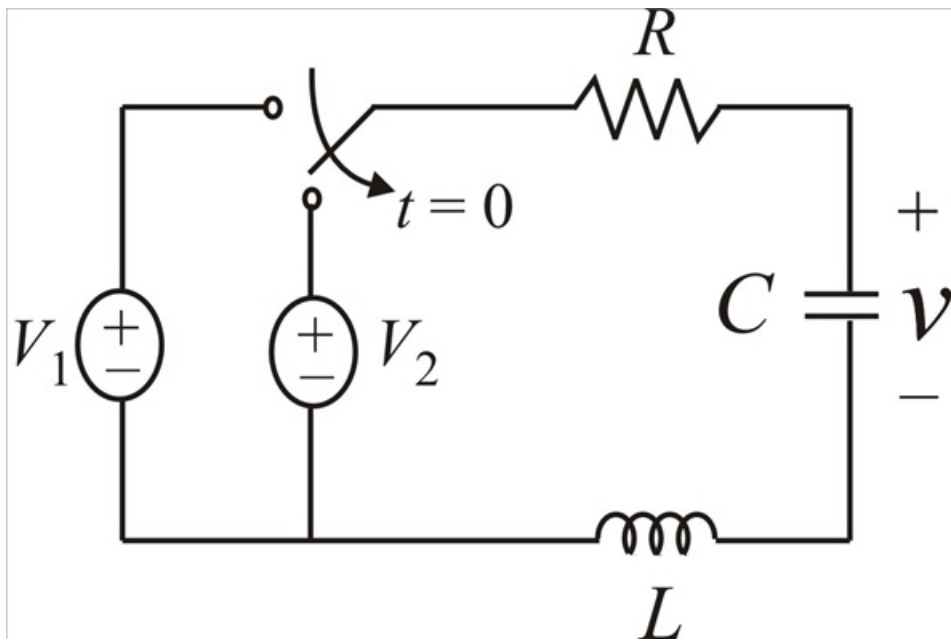
Figure 8.83:



Step-by-step solution

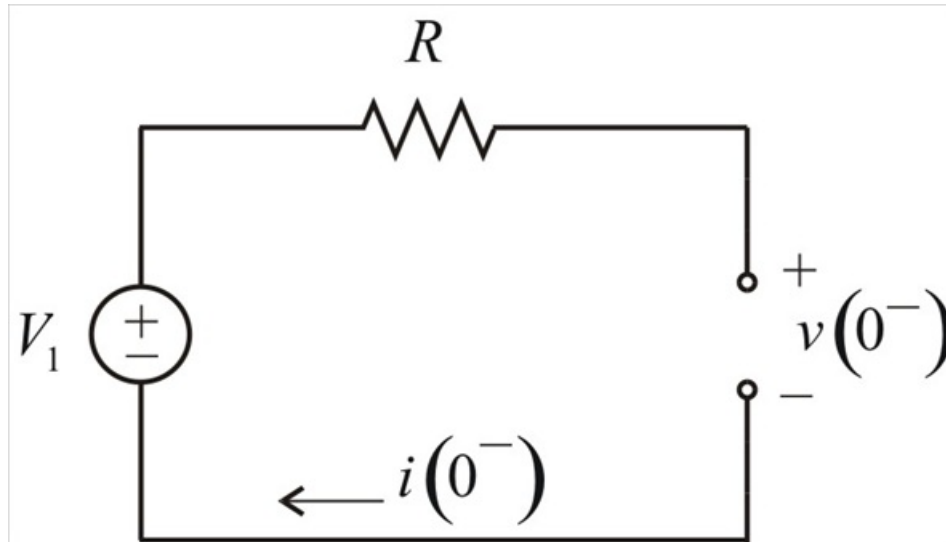
Step 1 of 9

The given circuit is



Step 2 of 9

For $t < 0$, the switch is at the top position and the circuit is in steady state condition, hence the capacitor can be replaced by an open circuit and the inductor by a short circuit,



Step 3 of 9

The voltage across the capacitor is,

$$v(0^-) = V_1$$

The initial current through the inductor is zero due to the open circuit,

$$i(0^-) = 0\text{A}$$

Step 4 of 9

Due to the continuity of the capacitor voltage and inductor current,

$$v(0^+) = v(0^-)$$

$$v(0^+) = V_1$$

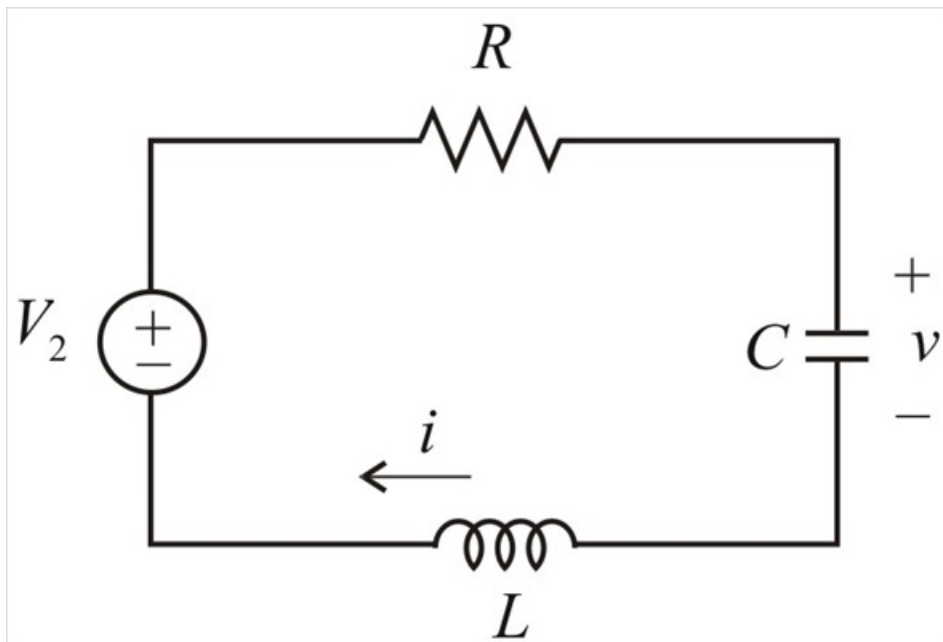
$$i(0^+) = i(0^-)$$

$$i(0^+) = 0\text{A}$$

Step 5 of 9

For $t > 0$, the switch is moved to the bottom position and the equivalent circuit is,

Step 6 of 9



Step 7 of 9

It is a series RLC circuit,

$$\alpha = \frac{R}{2L}$$

$$\omega_o = \frac{1}{\sqrt{LC}}$$

The roots are,

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_o^2}$$

Step 8 of 9

There are three types of response,

1. $\alpha > \omega_o$, over damped response

$$v(t) = V_{ss} + A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Where V_{ss} is the steady state response, i.e., the final value of the capacitor voltage.

Here, $v(\infty) = V_2$

$$\therefore V_{ss} = V_2$$

$$v(t) = V_2 + A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Step 9 of 9

2. $\alpha = \omega_o$, critically damped response.

$$v(t) = V_{ss} + (A_1 + A_2 t) e^{-\alpha t}$$

3. $\alpha < \omega_o$, under damped response

$$v(t) = v_{ss} + (B_1 \cos \omega_d t + B_2 \sin \omega_d t) e^{-\alpha t}$$

Where $\omega_d = \sqrt{\omega_o^2 - \alpha^2}$

The constants A_1, A_2, B_1 and B_2 can be determined using the initial conditions.